When Is the Vector Enough? Frozen Residual Readouts versus Answer-Token Scoring

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Abstract

When does a language model's frozen residual already support the answer, and when does answer-token scoring add value? We compare a small supervised classifier on final-token residual states with zero-/few-shot closed-set answer-token scoring across six open-weight models on BoolQ, RuleTaker, and ARC-Challenge. The methods use the same frozen backbone but different task supervision and prompt lengths. Their relative performance is structured rather than universal: the final-layer readout is higher than requested eight-shot scoring in five of six RuleTaker cases and two of six BoolQ cases, whereas scoring is higher in all six ARC cases. On ARC, the scorer's advantage falls from 10.6 points for Qwen3-0.6B to 0.4 points for Qwen3-8B as the readout becomes more predictive.

Existing BoolQ small-budget refits qualify the result: the full-training readout advantage is not established as supervision-independent. Intermediate layer sweeps show that answer-relevant information is often accessible before the final layer, although evaluation-selected maxima do not establish held-out gains or early exit. The scorer is the stored product-of-token-probabilities heuristic, not marginalized answer likelihood or free-form generation. The contribution is therefore a protocol-specific regime map: readouts are often competitive on rule- and passage-grounded tasks, while scoring is most useful when the residual readout is still weak.

1 Introduction

A frozen language model offers more than one interface for classification. One can use its answer-token outputs, or train a separate classifier on its internal representations. These interfaces need not give the same accuracy: the output head was trained for next-token prediction, while a supervised readout is fitted to the target task's labels. A useful empirical question is how the two procedures compare under explicitly stated training and inference budgets, rather than whether an answer is literally present in a vector.

We compare a one-hidden-layer MLP on final-token residual states with zero-/few-shot answer-token scoring. Both use a frozen backbone and one prompt forward pass per example, but they do not use identical prompts or equal amounts of task-specific supervision. In the main experiments, the classifier is fitted on thousands of labeled examples; the scorer receives a small number of labeled demonstrations in its prompt. The scorer is allowed a longer context, which can preserve more passage text and demonstrations but does not compensate in a controlled way for the classifier's training labels.

Our contribution is a cross-model, cross-task map of when these two readouts separate, together with a supervision-budget qualification and layerwise readouts. This is not a supervision-matched superiority result or a test of generated iterative reasoning. The main observations are:

1. **The loop's value is regime-dependent.** The final-layer classifier is numerically higher than the fixed eight-shot scorer in five RuleTaker cases and two BoolQ cases, while the scorer is higher in all six ARC cases. The largest scorer gaps occur where the readout is weakest, and the ARC gap narrows as readout accuracy rises (Table 1).

2. **Supervision changes the comparison.** In the existing BoolQ refits, matching the number of classifier training labels to the requested number of demonstrations does not preserve the full-training classifier's advantage (Table 4). These artifacts match declared counts, not independently verified example identities or retained prompt counts.

3. **Answer-relevant information is often accessible before the final layer.** Several intermediate layers support higher readout accuracy than the final layer. These evaluation-selected maxima do not establish held-out gains from layer selection or justify stopping the backbone early (Table 2).

The interpretation is further limited by the implemented scoring heuristic: its aggregation of alternative tokens differs from probability marginalization. We describe the exact rule below and retain its measured accuracies without presenting it as the strongest available output baseline.

2 Setup and Method

2.1 Supervised residual readout

Let $h_T \in \mathbb{R}^d$ denote the last nonpadding token's output from the final transformer block of a frozen model, before subsequent model-level normalization or output processing. For DeepSeek, grouped trailing output dimensions are flattened into a feature vector, giving an effective input width of 16384 (the other model widths range from 1024 to 4096). The prompt includes the question and, where supplied by the task, the passage. Gold answers are not appended to the readout's input. Prompts are left-truncated to 384 tokens, retaining the prompt suffix and trailing answer slot, but not necessarily the complete question. We standardize features using training-set means and standard deviations and fit a width-128, one-hidden-layer ReLU MLP. The implementation uses full-batch Adam with learning rate 10^{-3} for 150 epochs, with binary or multiclass cross-entropy as appropriate. The backbone is never updated; the MLP is nevertheless a task-supervised model.

The main readout statistic is the mean accuracy over four training seeds, with their standard deviation. It is neither the accuracy of the best seed nor that of a seed-vote ensemble. Logistic-regression readouts and other feature transformations are secondary analyses; selection of regularization or projection dimension on evaluation accuracy is exploratory.

2.2 Implemented answer-token scorer

The baseline processes a task prompt with zero or k balanced demonstrations from the training split. Demonstrations are selected in class round-robin order from the shuffled training rows; selection is not averaged over different exemplar sets or orders. Both methods choose from the same closed answer set: Yes/No for BoolQ and RuleTaker, and A–D for ARC. The baseline does not generate intermediate reasoning or a free-form answer.

For each answer class c, the implementation constructs a token-ID set S_c from bare and space-prefixed answer variants tokenized after `Answer:`. It slices off the number of tokens in the standalone prefix, unions the remaining IDs, and removes IDs shared by classes. At the last prompt token, it

computes

$$s_c(x) = \sum_{t \in S_c} \log p_\theta(t \mid x), \qquad \hat{y} = \operatorname*{argmax}_c s_c(x). \tag{1}$$

Thus it ranks the *product* of the individual token probabilities, not the sum of probability mass for mutually exclusive answer variants. Marginalization over single-token alternatives would instead use $\log \sum_{t \in S_c} p_\theta(t \mid x)$. Collecting IDs from a multi-token variant also does not compute that variant's sequence likelihood.

A local tokenizer audit found two IDs per Yes/No class for each of the six model tokenizers. On ARC, Mistral and Granite have two IDs per class; the Qwen and DeepSeek tokenizers yield one retained ID per class. In the latter cases, bare letter variants change the prefix tokenization and contribute no suffix IDs under the implemented slicing rule. Consequently, even where sum and marginalization coincide for the retained singleton, the procedure does not establish coverage of all intended answer variants.

All reported loop accuracies use this existing heuristic. Its numerical difference from a corrected scorer is unmeasured. The pipeline saves aggregate accuracies and, on fresh runs, class predictions rather than the token probabilities needed to rescore offline. These results should not be read as comparisons against standard answer-likelihood scoring generally.

2.3 Supervision, context, and reporting

The main readouts use 9427 training labels on BoolQ, 10000 on RuleTaker, and 1000 on ARC. The ARC filter leaves 1117 four-choice training rows and 1165 test rows; the reported training set is a seed-0 subsample of 1000 rows. We report zero- and eight-shot scorer arms separately, without selecting the stronger arm for the main comparison. This is a reporting choice for the present descriptive analysis, not a preregistered experimental design. Additional BoolQ runs requested up to 64 demonstrations.

The scorer's token cap is 8192 for Qwen BoolQ runs and 2048 otherwise, compared with the readout's 384. Left truncation can remove passage content, complete demonstrations, or parts of demonstrations. The stored results do not measure per-row truncation or the number of retained complete demonstrations. We therefore use *requested shots*, not effective shots. Equal question identities and answer sets do not imply equal token-level inputs or supervision.

Table 1 reports seed means and SDs for the classifier and single accuracies for the scorer. SD describes training-seed variation on the fixed evaluation set, not a confidence interval for population performance. Historical paired tests used separately fitted four-seed vote predictions; they do not directly test the displayed seed-mean differences. We do not attach those tests to the present descriptive tables. Small numerical differences are not evidence of equivalence, and no count of significant wins or corrected discoveries is claimed.

2.4 Models and datasets

The models are Qwen3-0.6B, Qwen3-4B, Qwen3-8B, Mistral-7B-v0.3, Granite-3.1-8B-base, and DeepSeek-V4-Flash (the stored model slug is `DeepSeek-V4-Flash-0731`). These are six models from four families, including dense and mixture-of-experts architectures, not a controlled architecture or parameter-count ablation. Reported backbone runs use batch size 8; DeepSeek uses the local quantized checkpoint loading path. Training-seed SD does not include variation from backbone numerical execution.

BoolQ [9] uses the `google/boolq` training split and its 3270-row validation split. RuleTaker [11] uses `tasksource/ruletaker`, with 10000 sampled training rows and 4000 sampled rows from

Table 1: Fixed final-layer readout and closed-set loop scores (accuracy). Readout is the stored `last.mlp` mean ± seed SD (four seeds; SD uses divisor four, not a confidence interval). `loop.zero` and `loop.8` are zero-shot and requested 8-shot accuracies under the manuscript's product-of-token-probabilities heuristic, not marginalized answer probabilities or free-form generation. Δ = readout mean − `loop.8`, with no baseline selection or significance testing. Artifact-declared train/evaluation/loop counts: BoolQ 9427/3270/3270; RuleTaker 10000/4000/4000; ARC-Challenge 1000/1165/1165. BoolQ Mistral/Granite readouts use sweep files and loops use full-evaluation reruns, not the sweeps' 405-row loops. Readout token cap is 384; loop cap is 8192 for BoolQ Qwen and 2048 otherwise. These are different input and supervision protocols, not a matched-label-budget comparison; requested demonstrations are not measured retained counts.

Task	Model	Readout mean ± SD	`loop.zero`	`loop.8`	Δ vs. 8-shot
BoolQ	Qwen3-0.6B	0.746 ± 0.004	0.631	0.645	+0.101
	Qwen3-4B	0.856 ± 0.002	0.855	0.865	−0.009
	Qwen3-8B	0.878 ± 0.002	0.862	0.876	+0.001
	Mistral-7B	0.841 ± 0.002	0.824	0.863	−0.022
	Granite-3.1-8B	0.854 ± 0.001	0.800	0.865	−0.011
	DeepSeek-V4-Flash	0.896 ± 0.001	0.888	0.906	−0.009
RuleTaker	Qwen3-0.6B	0.684 ± 0.004	0.593	0.580	+0.103
	Qwen3-4B	0.782 ± 0.003	0.678	0.690	+0.092
	Qwen3-8B	0.782 ± 0.003	0.678	0.736	+0.046
	Mistral-7B	0.682 ± 0.005	0.522	0.571	+0.111
	Granite-3.1-8B	0.823 ± 0.002	0.564	0.703	+0.119
	DeepSeek-V4-Flash	0.797 ± 0.002	0.600	0.813	−0.016
ARC-Challenge	Qwen3-0.6B	0.492 ± 0.007	0.502	0.597	−0.106
	Qwen3-4B	0.841 ± 0.001	0.826	0.882	−0.041
	Qwen3-8B	0.909 ± 0.002	0.902	0.913	−0.004
	Mistral-7B	0.728 ± 0.003	0.750	0.779	−0.051
	Granite-3.1-8B	0.708 ± 0.006	0.732	0.790	−0.082
	DeepSeek-V4-Flash	0.951 ± 0.001	0.961	0.959	−0.008

the official *test* split. The artifacts call the latter `val`; this is an evaluation-set name, not the official validation split. ARC-Challenge [10] uses `allenai/ai2_arc`, retaining four-choice items, with 1117 filtered training rows and 1165 test rows. The reported 1000-row training set is a seed-0 subsample. No retrieval corpus is supplied for ARC, but the question and choices themselves remain part of the input.

3 Results

Figure 1 visualizes the fixed final-layer and separate scorer arms reported in Table 1.

3.1 BoolQ: full-training and small-budget readouts

Table 1 shows that BoolQ is a mixed but informative regime. Qwen3-0.6B has a classifier mean of 0.746 versus 0.645 for eight-shot scoring, while Qwen3-8B is nearly even at 0.878 versus 0.876. The other models favor scoring numerically, so the result is not a universal readout win. It is nevertheless consistent with the broader map: once both channels are strong, their difference is small; where the residual readout is weaker, the scorer can retain an advantage. Small differences are not interpreted as statistical ties.

Table 2: Descriptive readout placement sweeps (accuracy). Selected layer maximizes the stored MLP seed mean over recorded sweep entries (including the final layer where swept). Layer indices are zero-based. Selected and final means come from the same sweep JSON; the final column uses that sweep file's `last.mlp`, not a value imported from Table 1. ARC Qwen3-0.6B has no L27 sweep entry; its final mean is stored separately in the same file. These evaluation-selected maxima are optimistic, not held-out gains; no significance tests are reported. Sweep train/evaluation counts are 9427/3270 (BoolQ), 10000/4000 (RuleTaker), and 1000/1165 (ARC-Challenge). RuleTaker uses the separate 10k/4k sweep files because the headline 10k/4k files contain no sweep. Source runs and protocols may differ from the headline sources even when displayed final means coincide.

Task	Model	Selected layer	Selected mean	Final layer	Final mean
BoolQ	Qwen3-0.6B	L18	0.761	L27	0.746
	Qwen3-4B	L24	0.868	L35	0.856
	Qwen3-8B	L30	0.889	L35	0.878
	Mistral-7B	L16	0.863	L31	0.841
	Granite-3.1-8B	L20	0.868	L39	0.854
	DeepSeek-V4-Flash	L36	0.900	L42	0.896
RuleTaker	Qwen3-0.6B	L18	0.720	L27	0.684
	Qwen3-4B	L24	0.796	L35	0.782
	Qwen3-8B	L24	0.797	L35	0.782
	Mistral-7B	L16	0.732	L31	0.682
	Granite-3.1-8B	L20	0.845	L39	0.823
	DeepSeek-V4-Flash	L29	0.807	L42	0.797
ARC-Challenge	Qwen3-0.6B	L26	0.500	L27	0.492
	Qwen3-4B	L29	0.856	L35	0.841
	Qwen3-8B	L35	0.909	L35	0.909
	Mistral-7B	L20	0.741	L31	0.728
	Granite-3.1-8B	L39	0.708	L39	0.708
	DeepSeek-V4-Flash	L42	0.951	L42	0.951

The existing requested-shot sweep qualifies the Qwen3-8B comparison: its full-training readout (0.878) is below the recorded 16-, 32-, and 64-shot scorer arms (0.886, 0.880, and 0.883, rounded). For Qwen3-0.6B, by contrast, the readout (0.746) exceeds every recorded scorer arm; its scorer rises from 0.631 at zero shots to 0.715 at both 32 and 64 shots (rounded), reducing the readout's lead to about 3.0 percentage points. Similar 32- and 64-shot accuracies in this one protocol do not establish saturation: context truncation and the single exemplar ordering are not separately controlled.

Table 4 reports existing cache-refit artifacts for 8, 16, 32, and 64 training labels on the three Qwen models. They contain both first-n and class-balanced refits. In all 12 model–budget settings, both refit means are below the corresponding requested-shot scorer. For example, the 0.6B class-balanced 64-label readout scores 0.640, compared with 0.715 for requested-64-shot scoring and 0.746 for the full-training readout. At 64 labels the corresponding balanced readouts are 0.855 and 0.849 for 4B and 8B, versus scorer accuracies of 0.867 and 0.883.

These comparisons do not prove that label count explains the whole full-training difference. They do show that the observed advantage is not established as supervision-independent. The refit metadata declares label counts and subset protocols, but the exact balanced subset identities and the cache-refit generator have not been recovered. We treat these as documented existing measurements, not a fully reproduced identical-example experiment.

Figure 2 shows these refits alongside the requested-shot curves and the full-training readout

Table 3: Descriptive control readouts (accuracy). All entries use the headline readout JSONs and train/evaluation counts of Table 1. Shuffle is the stored mean for an MLP refit on permuted training labels (`last.mlp.shufl`); context and final are `ctx.mlp` and `last.mlp` seed means. RP max, its dimension, perm, and noise are the stored `last.randproj` values: linear probes on Gaussian random projections, feature-column-permuted inputs, and Gaussian noise inputs. RP dimension is selected on evaluation accuracy from 64 and 256 using three-seed means; perm and noise use the selected dimension. These are descriptive measurements, not leakage tests or causal evidence.

Task	Model	Shuffle	RP max	RP d	RP perm	RP noise	Context	Final
BoolQ	Qwen3-0.6B	0.535	0.734	256	0.725	0.601	0.667	0.746
	Qwen3-4B	0.542	0.861	256	0.859	0.604	0.717	0.856
	Qwen3-8B	0.528	0.874	256	0.874	0.607	0.739	0.878
	Mistral-7B	0.537	0.810	256	0.803	0.607	0.685	0.841
	Granite-3.1-8B	0.532	0.848	256	0.849	0.607	0.679	0.854
	DeepSeek-V4-Flash	0.522	0.889	256	0.891	0.602	0.678	0.896
RuleTaker	Qwen3-0.6B	0.498	0.657	256	0.663	0.502	0.532	0.684
	Qwen3-4B	0.516	0.748	256	0.752	0.490	0.619	0.782
	Qwen3-8B	0.506	0.760	256	0.755	0.502	0.634	0.782
	Mistral-7B	0.503	0.656	256	0.646	0.502	0.521	0.682
	Granite-3.1-8B	0.504	0.792	256	0.785	0.502	0.599	0.823
	DeepSeek-V4-Flash	0.509	0.768	256	0.759	0.503	0.526	0.797
ARC-Challenge	Qwen3-0.6B	0.278	0.470	64	0.470	0.254	0.278	0.492
	Qwen3-4B	0.269	0.802	64	0.803	0.255	0.349	0.841
	Qwen3-8B	0.287	0.869	256	0.867	0.251	0.399	0.909
	Mistral-7B	0.259	0.679	256	0.678	0.251	0.277	0.728
	Granite-3.1-8B	0.284	0.598	256	0.588	0.251	0.279	0.708
	DeepSeek-V4-Flash	0.269	0.942	256	0.940	0.236	0.439	0.951

reference.

3.2 RuleTaker and ARC-Challenge

On the 4000-row RuleTaker evaluation subset, the final-layer classifier mean is higher than the eight-shot scorer for five models, with differences from approximately +4.6 to +11.9 percentage points. This is the clearest readout-favorable regime in the study: a task-grounded classifier on the frozen residual substantially improves on the stored answer-token decision rule for several backbones. DeepSeek's classifier is lower by about 1.6 points, showing that the pattern is not universal. Zero-shot scoring is higher than eight-shot scoring for Qwen3-0.6B; against zero-shot its readout difference is about +9.0 rather than +10.3 points. The main table keeps both scorer arms visible rather than selecting one using evaluation accuracy.

The stored RuleTaker `depth` field is parsed from the release's configuration name, not measured from a proof for each question. In particular, `depth-3`, `depth-3ext`, and `depth-3ext-NatLang` contribute to the same numeric group. We do not interpret the historical configuration-stratified results as an experiment isolating question-level deduction depth, nor infer a serial reasoning threshold from them.

On ARC, the eight-shot scorer is numerically higher in all six cases. This is the clearest scorer-favorable regime, but its size is organized by readout competence: the gap is about 10.6 points for Qwen3-0.6B, 4.1 points for Qwen3-4B, and 0.4 points for Qwen3-8B. Granite remains an exception at an 8.2-point gap despite a stronger model, while DeepSeek is nearly even. The pattern is an

Table 4: BoolQ label-budget descriptions (accuracy). First-n and class-balanced columns are stored refit seed means from `budget` and `budget.balanced` in the same model's `samek` JSON; loop values are that JSON's `loop.k`. $n = k$ compares artifact-declared readout label counts with requested demonstration counts, not measured retained demonstration counts. Exact subset identity is unavailable, so equal counts do not establish identical labeled examples. The declared training pool has 9427 rows; readout and loop evaluation counts are both 3270. Readout/loop token caps are 384/8192. No significance tests are reported.

Model	$n = k$	First-n refit	Class-balanced refit	`loop.k`
Qwen3-0.6B	8	0.622	0.612	0.645
	16	0.607	0.594	0.669
	32	0.625	0.619	0.715
	64	0.660	0.640	0.715
Qwen3-4B	8	0.850	0.851	0.865
	16	0.828	0.829	0.862
	32	0.839	0.844	0.861
	64	0.853	0.855	0.867
Qwen3-8B	8	0.826	0.825	0.876
	16	0.821	0.812	0.886
	32	0.834	0.845	0.880
	64	0.850	0.849	0.883

empirical association, not a causal size law: readout accuracy and readout-minus-scorer share a term mathematically, and the readout has only 1000 training labels here versus 9427 on BoolQ and 10000 on RuleTaker. The 1000 ARC rows are most of the 1117 filtered training rows, so historical 1117-label Qwen results provide only corroborating, not drop-in, evidence about the endpoint. Qwen3-0.6B's readout accuracy of 0.492 is well above the 0.25 uniform-guess baseline for four choices; its lower accuracy does not show that no other classifier could extract useful information.

3.3 Exploratory layer sweeps

Table 2 summarizes the recorded layer sweeps using the highest seed-mean accuracy among the swept layers and the final-layer readout from the same source run. The result is not merely that the final state is readable: intermediate states often support stronger task readouts. For example, the recorded RuleTaker Mistral sweep has a selected mean of 0.732 at layer 16 versus 0.682 at the final layer. The selected layer varies across models and tasks; several ARC maxima occur at the final layer.

Selection uses the same evaluation labels on which accuracy is reported, so the selected maxima are optimistic estimates of a layer-selection procedure's generalization. A paired test at the selected layer, followed only by correction across the 18 cells, would not account for the within-cell layer search. We therefore report neither selection-unadjusted significance nor a claim that intermediate layers never lose. These sweeps describe predictive readouts, not the behavior of a backbone stopped early.

Figure 3 displays all recorded MLP sweep points rather than highlighting evaluation-selected maxima.

3.4 Control readouts

Table 3 gives label-shuffle, random-projection, noise, and mean-pooled-context results. Context pooling averages all retained nonpadding prompt tokens, not just passage tokens. Shuffling training

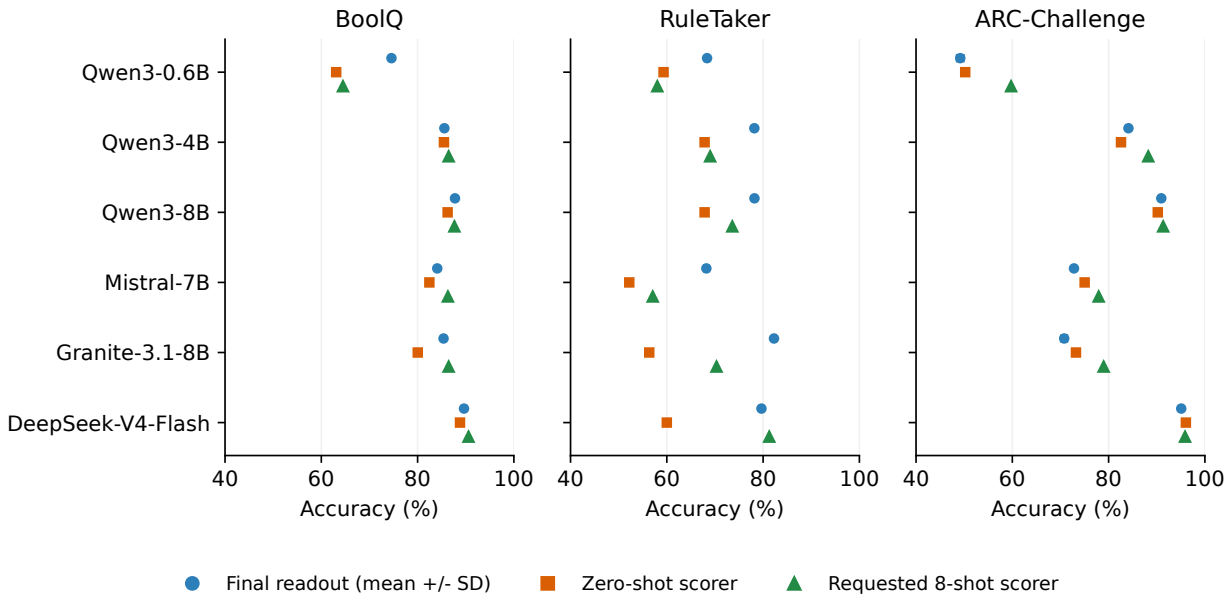


Figure 1: Cross-model accuracy under different supervision protocols. Readout points are final-layer four-seed means; bars show seed SD, not confidence intervals. Scorer points retain both zero-shot and requested eight-shot arms of the implemented token-probability heuristic. Readouts use 9427 BoolQ, 10000 RuleTaker, or 1000 ARC training labels; scorers do not fit task-specific weights. Prompt lengths and token caps differ (Table 1). No significance tests are shown.

labels generally reduces predictive accuracy, but this is not a dataset-overlap audit and does not rule out leakage. On imbalanced datasets, uniform guessing and majority-class prediction have different expected accuracies; the BoolQ validation majority is 0.6217.

Gaussian projections retain substantial predictive accuracy at the tested dimensions (64 or 256, reported per row in Table 3). The reported dimension is selected on evaluation accuracy; these are descriptive results. An isotropic Gaussian projection is distributionally invariant to a coordinate permutation, so similar permuted-coordinate results cannot establish that no privileged subspace exists. Finally, the tested mean-pooled-context MLPs are less accurate than the final-token MLPs. This is a comparison of feature summaries, not evidence that a verdict is causally computed at the final token or that passage-level shortcuts are absent.

4 Discussion and Limitations

The central result is a regime map, not a universal winner. A task-specific classifier on a frozen residual is often competitive with the stored closed-set scorer and is substantially better on most of the RuleTaker models. Conversely, scoring has its clearest advantage on ARC when the residual readout is weak, and that advantage narrows as the readout becomes more accurate. The comparison changes both the readout and the training objective, supervision budget, and prompt, so these patterns should not be read as supervision-independent superiority or as proof that the residual alone contains a model belief.

Scorer validity. The product-of-token-probabilities heuristic is not standard marginal answer likelihood. Correcting tokenization and aggregation could change the baseline rankings and numerical conclusions; the stored class predictions cannot quantify that change. All comparative claims are conditional on the implemented scorer, not claims about the best use of the model's output

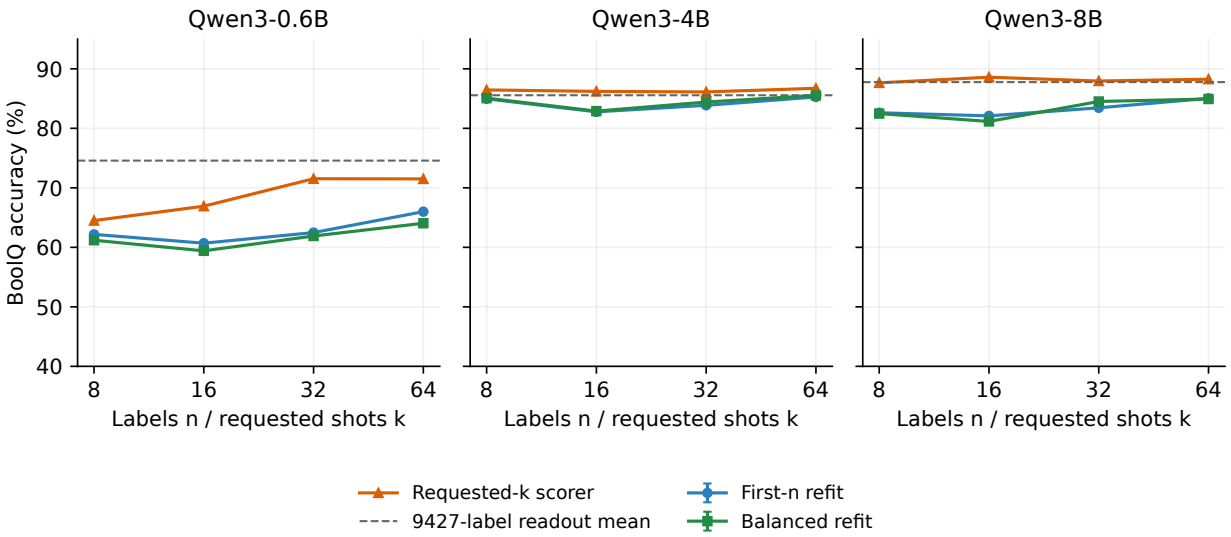


Figure 2: BoolQ supervision-budget descriptions. At $n = k \in \{8, 16, 32, 64\}$, readout curves show first-n and class-balanced refit means with seed SD; scorer curves show requested-k-shot accuracy. Dashed lines mark the 9427-label readout mean, not a matched-budget arm. Equal counts do not certify identical examples or retained demonstrations; readout/scorer token caps are 384/8192. Lines connect recorded settings, not fitted scaling laws. Sources match Table 4.

distribution.

Supervision is part of the result. Headline label budgets are unequal. Stored first-n readouts at $n = 64$ exist for all 18 headline cells and are below the corresponding eight-shot scorer in 17 of 18 cells, including all six RuleTaker cells. This is not matched-budget evidence: RuleTaker and ARC have no 64-shot scorer arm, and these stored refits are first-n, not class-balanced. The BoolQ class-balanced refits provide a complementary qualification for three models on one dataset, with incomplete subset provenance. Neither result establishes general sample efficiency. Effective demonstration counts, truncation rates, and sensitivity to exemplar selection and order remain unmeasured.

Evaluation and selection. This is a retrospective descriptive analysis of existing runs. Evaluation sets informed earlier layer, hyperparameter, and reporting choices; there is no new untouched test set. Seed SD does not capture dataset-sampling or prompt-selection uncertainty. The RuleTaker results cover a sampled subset rather than its full test set. Duplicate and train–evaluation overlap counts have not been audited here; the shuffle controls do not replace those checks.

Computation and mechanisms. Both main methods use a single prompt forward pass, with different lengths; neither generates intermediate reasoning tokens. The experiments therefore do not compare one-pass computation with iterative computation, locate a circuit-complexity boundary, or explain why a particular model favors one method. We make no latency, throughput, early-exit, or production-serving claim, and no comparison to fine-tuned specialists under matched resources.

Artifact coverage. Table sources are explicit in `scripts/build_descriptive_tables.py`, which regenerates the descriptive tables from stored JSONs. Sweep and headline entries are distinct in the source mapping only where separate files are used; some entries share a file or cached computation, so the tables do not provide a run-to-run variability estimate. Local full-size paired loop/readout predictions are incomplete for BoolQ Qwen3-0.6B, Qwen3-8B, and DeepSeek; scalar results exist for all 18 cells. Persisted best-seed heads, seed means, and separately fitted vote ensembles are different objects and should not be substituted for one another.

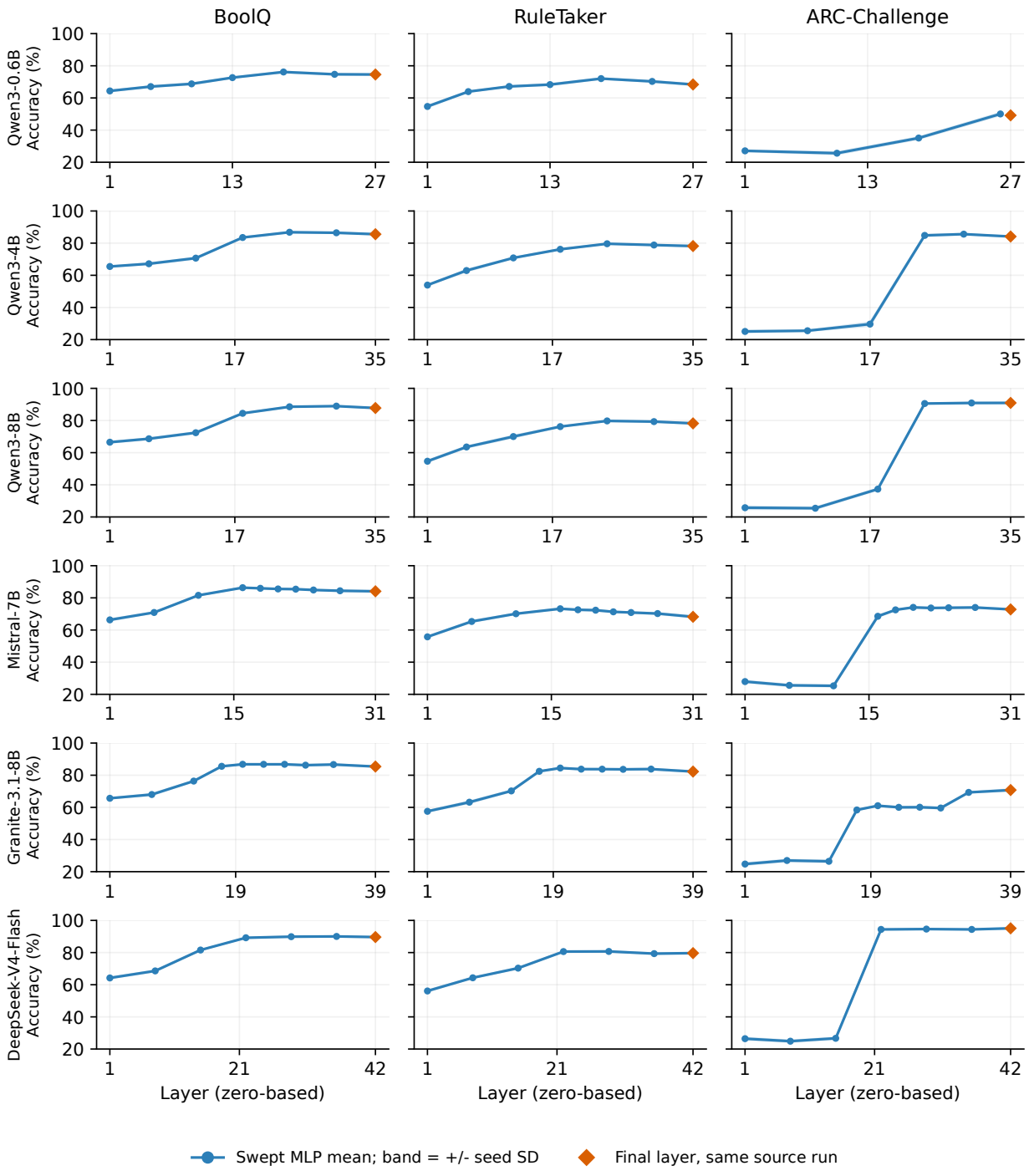


Figure 3: Exploratory layer sweeps. Curves connect recorded MLP seed means; bands show ± seed SD, not confidence intervals. Diamonds mark the final-layer readout from the same sweep JSON, including the separately stored ARC Qwen3-0.6B final L27. Sources match Table 2, not necessarily the headline runs. Layer indices are zero-based. No winning layer or significance is marked; evaluation-selected maxima are not held-out gains.

5 Related Work

Frozen representations for prediction. Prior work studies differences between hidden-state classifiers and prompted or token-level decisions [1, 4, 7, 8]. Hazenoot et al. [17] compare frozen-activation probes with prompted yes/no predictions for ESG concept measurement. Our study adds a cross-model, three-dataset comparison under the particular protocols described here, not a new claim that supervised probes can predict task labels.

Beliefs and correctness. CCS [5] studies unsupervised recovery of truth-related directions; other work examines belief timing and self-knowledge [3, 12, 18]. Correctness prediction, candidate reranking, and hidden-knowledge studies [6, 14, 19, 21] address related but distinct targets. Our classifier predicts gold task labels from a prompt representation; that operational distinction does not identify the model's beliefs or prove that its output hides an already-computed answer.

Layerwise readouts and adaptive computation. The logit lens [20] and tuned lens [2] map intermediate representations toward vocabulary predictions. Patchscopes [15] and representation engineering [26] provide additional tools for inspecting or intervening on representations. Our supervised layer sweeps measure a different readout objective. Early-exit methods [13, 23–25] evaluate stopping policies, which we do not implement. Likewise, continuous latent reasoning and looped transformers [16, 22] involve trained iterative computation absent from our main comparison.

6 Conclusion

Across the evaluated models and datasets, frozen residual readouts and closed-set answer-token scoring exhibit structured rather than universal differences. Readouts are higher in most RuleTaker cases and remain competitive on BoolQ, while scoring has its clearest advantage on ARC models whose residual readouts are weak. The ARC gap narrows as readout accuracy rises, suggesting that the loop is most useful when the residual does not yet provide an easily extractable task decision. Existing small-budget refits show that the full-training readout advantage depends partly on task supervision and should not be interpreted as supervision-independent superiority. Intermediate-layer sweeps further indicate that answer-relevant information is often accessible before the final layer, although they do not establish early exit. The contribution is a bounded empirical regime map, conditional on the stored scorer and evaluated protocols, rather than a claim that the loop is unnecessary or that every answer is already in the vector.

Data availability

The experiments use public releases of BoolQ [9], RuleTaker [11], and ARC-Challenge [10], with splits, subsampling, and filtering described in §2. Stored result JSONs and the descriptive table generator accompany the manuscript in the repository at `https://github.com/hanan-herz/residual-readout-bench`. No new model inference or classifier fitting was performed for this revision.

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